

Rishabh Agnihotri

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LinkedIn

GitHub

LeetCode

CodeChef

EDUCATION

Indian Institute of Information Technology, Bhopal

Jul 2023 – 2027

B.Tech, Electronics and Communication Engineering — CGPA **8.63**

Bhopal, India

Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

TECHNICAL SKILLS

Languages: C++, JavaScript, Python, SQL

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, Redux, Tailwind CSS, Socket.IO

Databases & Tools: MongoDB, PostgreSQL, MySQL, Redis, Git, GitHub, Docker, Postman, VS Code

Backend & Auth: REST APIs, JWT Authentication, OAuth 2.0, RBAC, Middleware, Bull Queue, WebSockets

Cloud & Deployment: Vercel, Render, MongoDB Atlas

Developer Tools & Practices: REST API Design, MVC Architecture, Git Workflow, CI/CD Basics, Debugging

PROJECTS

TrustTrade — Full Stack Escrow Marketplace

GitHub

Live

2026

Node.js, Express.js, MongoDB, Redis, Bull Queue, Razorpay, Socket.io, Passport.js, React.js, Google Gemini API

- Designed a secure **RESTful API** with **Node.js/Express.js** across **9 route modules** implementing a **7-state escrow lifecycle** with **RBAC middleware** (buyer/seller/admin), **rate limiting**, **Helmet.js** security headers, and **HMAC-SHA256 webhook signature verification** for Razorpay.
- Architected a **3-layer Redis** system — **payment idempotency locks** (5-min TTL), **webhook deduplication keys** (24-hr TTL), and **Bull Queue** for **7-day escrow auto-release** with **exponential backoff retry** — ensuring **exactly-once fund processing**; integrated **Google Gemini API** for **AI-powered product listing generation** and **automated dispute fraud detection**.
- Implemented **Google/GitHub OAuth 2.0** via **Passport.js** with **httpOnly refresh-token cookies**, **email OTP verification**, **refresh-token rotation** with replay attack detection, and **Socket.io JWT-authenticated WebSocket** handshakes for **real-time buyer-seller chat** and order notifications.

ChainVerify — Blockchain Document Verification Platform

GitHub

Live

2026

Node.js, Express.js, MongoDB, Solidity, Ethereum, Ethers.js, React

- Built **7 RESTful API endpoints** across **authRoutes**, **documentRoutes**, and **verifyRoutes** using **Express.js** with custom **JWT middleware** authenticating Bearer tokens, injecting users into **req.user**, and enforcing **ownership-level access control** so users can only verify or delete their own documents.
- Integrated **Ethers.js** as a singleton blockchain client (**JsonRpcProvider** + **Wallet signer**) with deployed **Solidity smart contracts** on **Sepolia testnet**, storing **SHA-256 hashes** on-chain for **tamper-proof verification**; optimised **gas costs** by checking **on-chain proof existence** before every write.
- Designed **MongoDB schemas** for User, Document, and Verification collections with **field-level validation**, **unique indexes**, **enum constraints**, and **soft-delete pattern** for **audit-trail preservation**; deployed on **Render + Vercel** with **MongoDB Atlas**.

FinTrack — Stock Portfolio Dashboard

GitHub

Live

2026

Node.js, Express.js, MongoDB, React, TailwindCSS, Recharts

- Architected **8 RESTful API endpoints** across authentication and portfolio modules using **JWT middleware** for stateless **session verification** and standardized **HTTP status codes** (200/201/400/401/404/500) for consistent **API response handling**.
- Designed a normalized **MongoDB data model** separating **portfolio holdings** from **transaction history** using **schema-level enums**, **Mongoose validators** (required constraints, uppercase transforms), **unique indexes** on user-stock pairs, and a **compound index** for efficient **sorted data retrieval**.
- Built a reusable **backend service layer** for **Finnhub API** integration with **60-second in-memory caching** (TTL-based), symbol normalization (NSE → Finnhub format), and **Promise.all()** for **concurrent price fetching** across all portfolio holdings.

ACHIEVEMENTS

- Secured **Rank 3** in Hack-O-Sprint 2.0 hackathon at IIIT Bhopal for Smart India Hackathon (SIH) 2024, building a working solution under strict time constraints.
- Achieved **3-Star on CodeChef** (highest rating **1642**) and **Global Rank 84** in CodeChef Starters 231 among thousands of competitors.
- Solved **600+ DSA problems** across LeetCode, CodeChef, and GeeksforGeeks; achieved a **LeetCode Contest Rating of 1800+** covering arrays, trees, graphs, and dynamic programming.